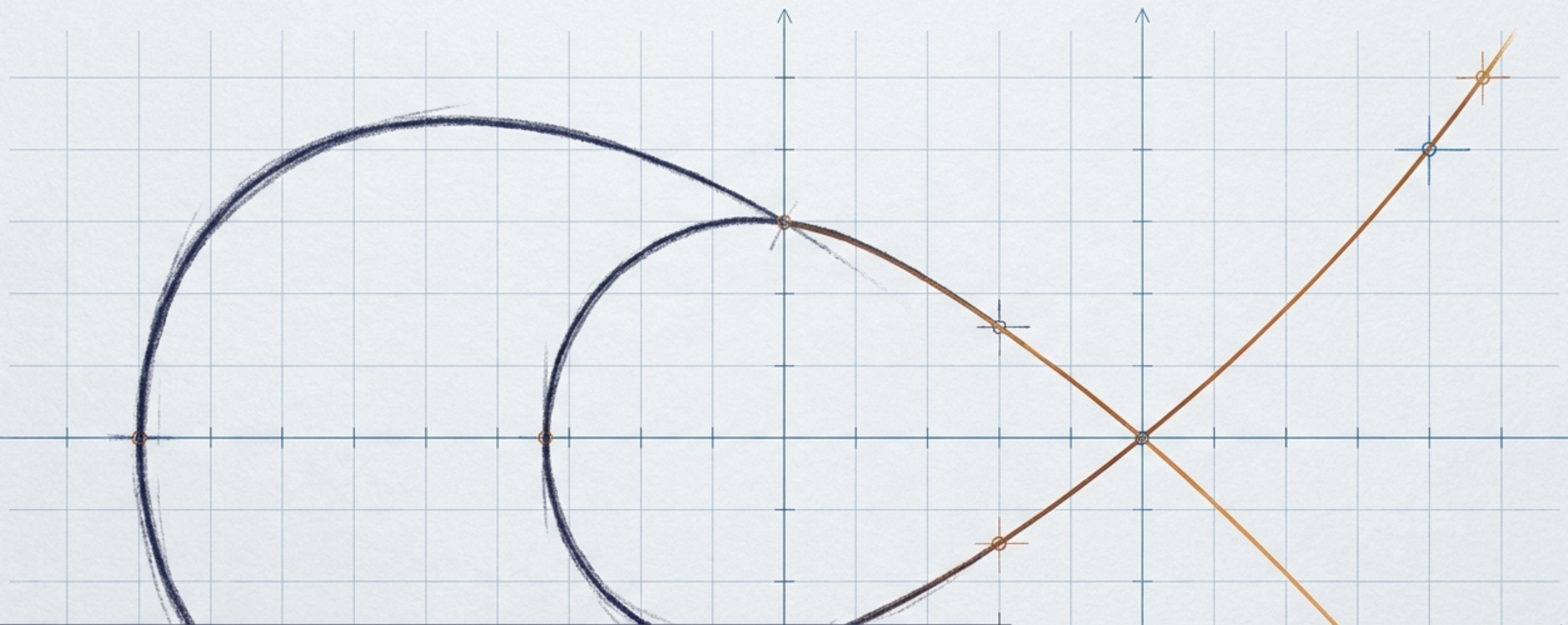




The image features a large grid with a horizontal and vertical axis. Two curves are plotted: a dark blue curve on the left and a brown curve on the right. They intersect at a point marked with a small circle and crosshairs. A straight line, colored orange-brown, passes through the intersection point and extends towards the top right. Several other points are marked along this line with small circles and crosshairs. The grid lines are light blue, and the axes have arrows at their ends.

The Rosetta Stone of Numbers

Decoding the Birch and Swinnerton-Dyer Conjecture:
From High-School Algebra to the 2026 AI Breakthrough

A Standalone Visual Explainer | October 2026

What is an Elliptic Curve?

The Equation

Despite the complex name, it is just a polynomial equation:

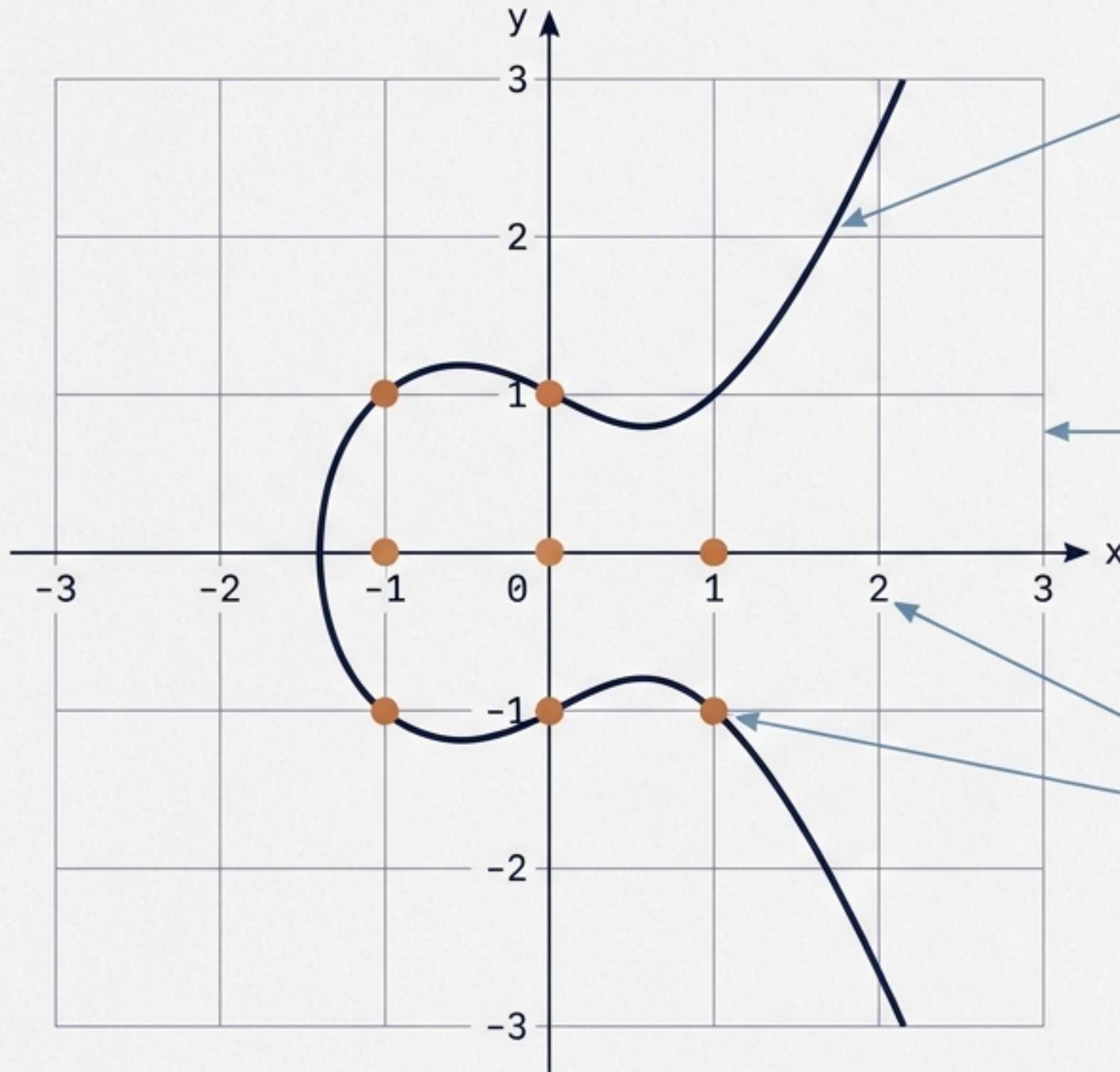
$$y^2 + y = x^3 - x.$$

The Shape

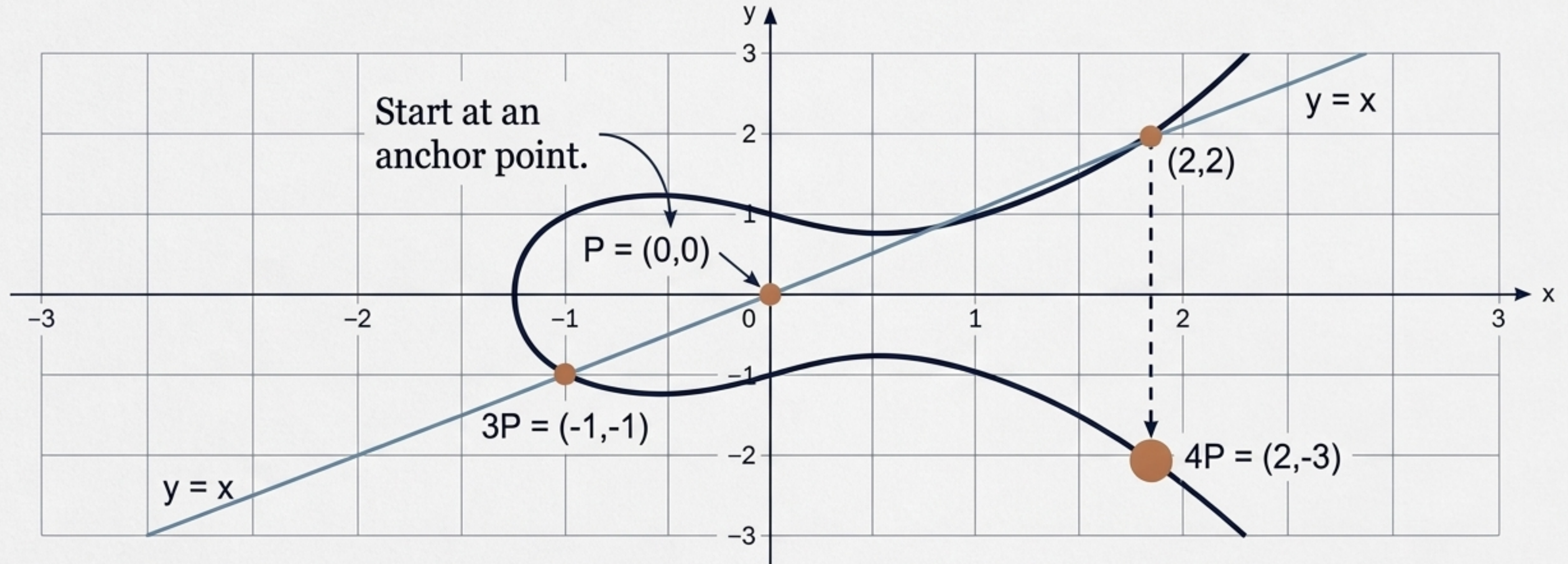
A smooth, symmetric curve on a standard x-y coordinate plane.

The Goal

Number theory seeks to find its **RATIONAL POINTS**—exact locations where both x and y are fractions or whole numbers, not endless decimals.



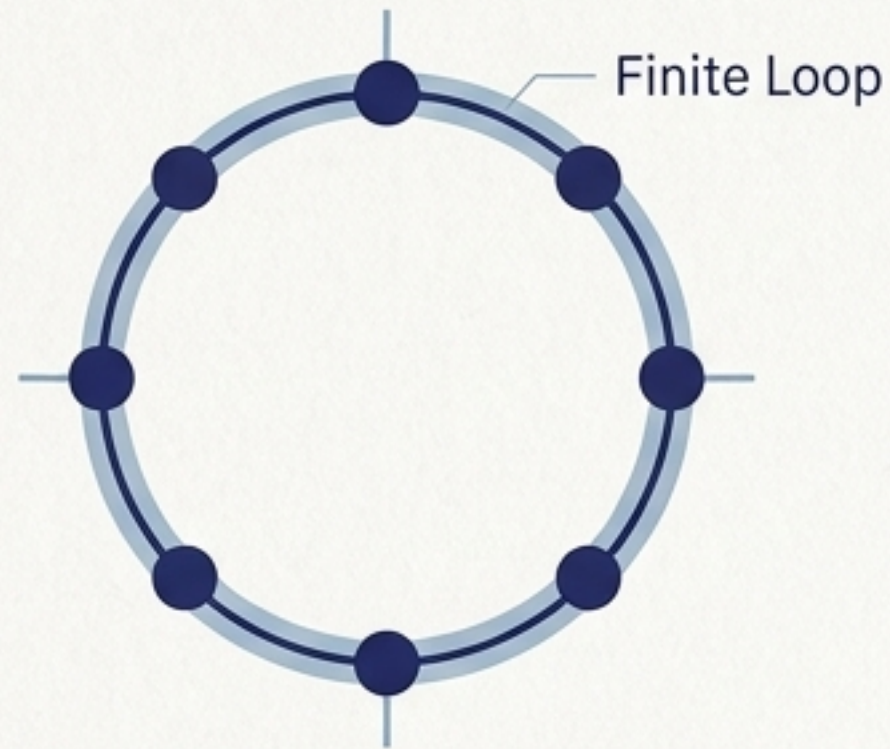
The Magic of the Chord Rule: Geometry as Arithmetic



By drawing lines through existing rational points, we magically discover new ones.
The points form a mathematical group where geometry dictates addition.

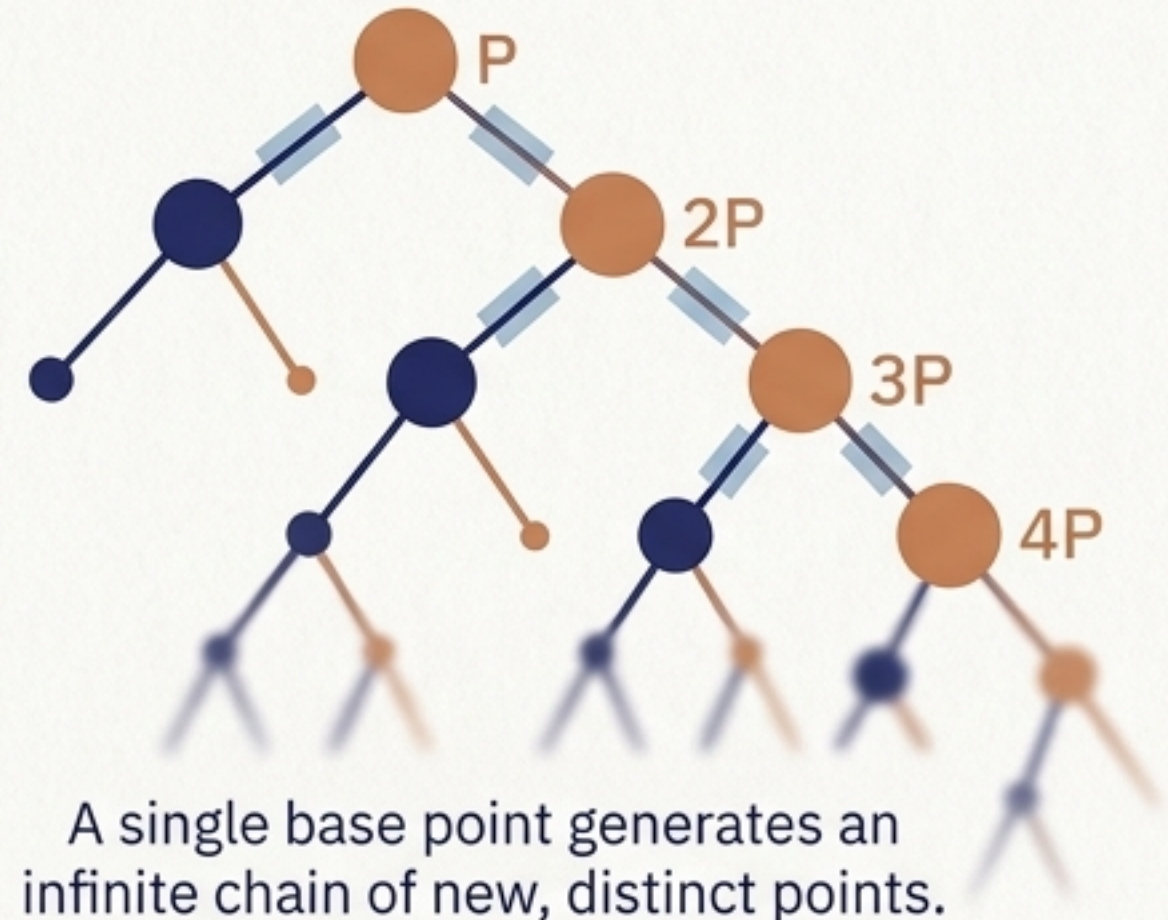
The Concept of Rank: Measuring Infinity

Rank 0



The curve has only a finite, closed loop of rational points.

Rank > 0



A single base point generates an infinite chain of new, distinct points.

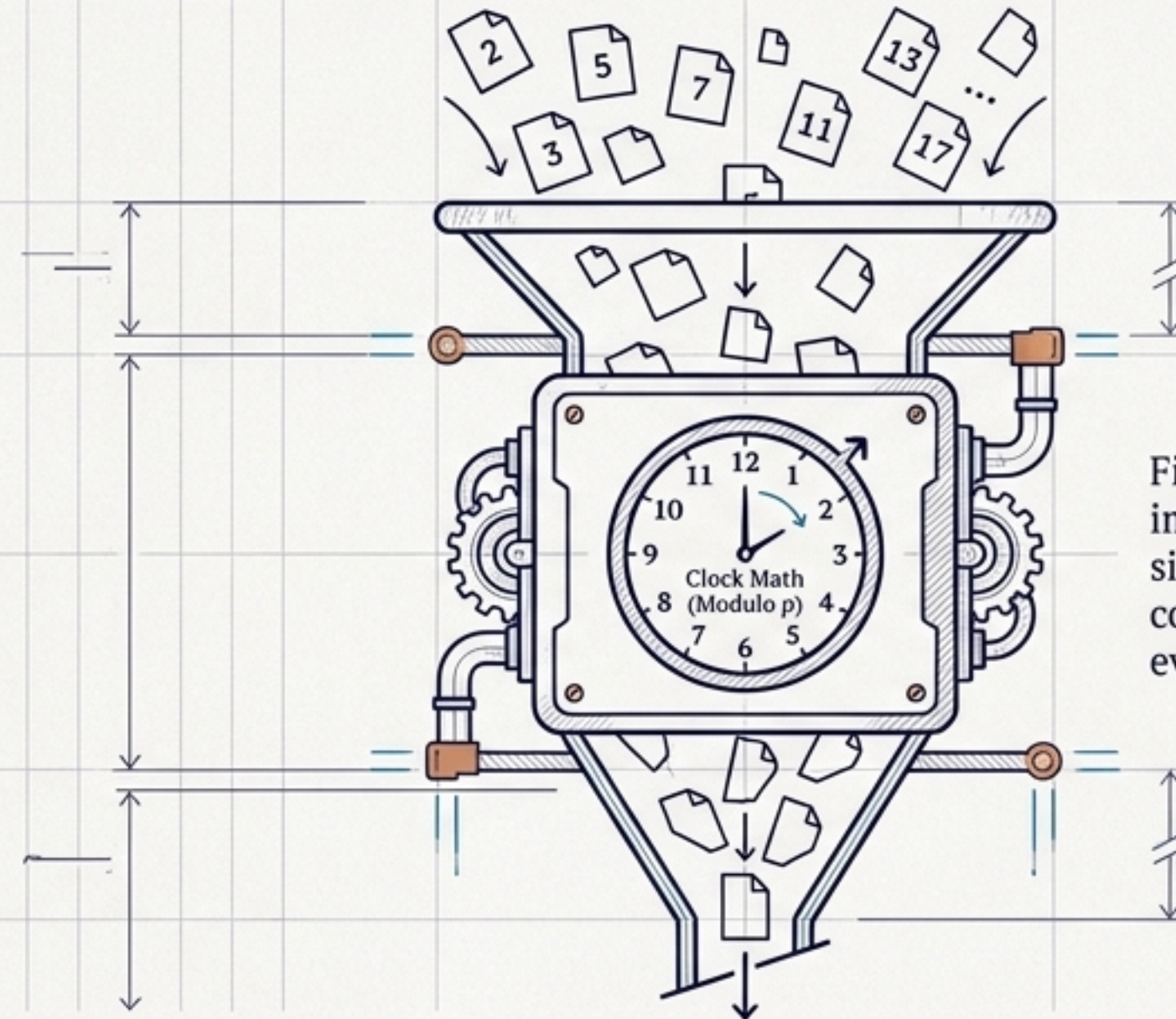
Rank counts the number of **INDEPENDENT** points of infinite order.
It is the core DNA of an elliptic curve.

The Arithmetic World: Counting Modulo Primes

Input

The Funnel

Output



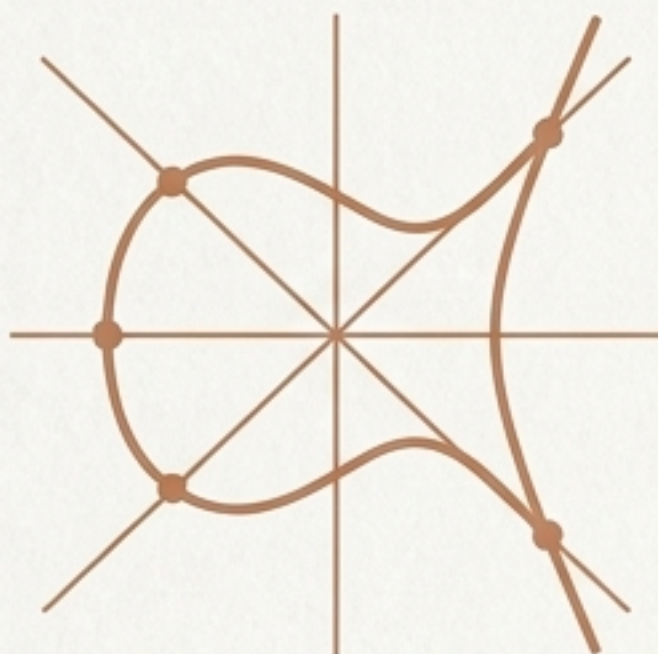
Finding exact fractions directly is impossibly hard. Instead, mathematicians simplify the curve into clock math and count the number of finite solutions for every prime number p .

The Hasse-Weil L-Function $L(E, s)$

↳ This L-function encodes the arithmetic behavior of the curve. The most critical information is hidden exactly at its center point: $s = 1$.

The Conjecture: The Ultimate Translation Bridge

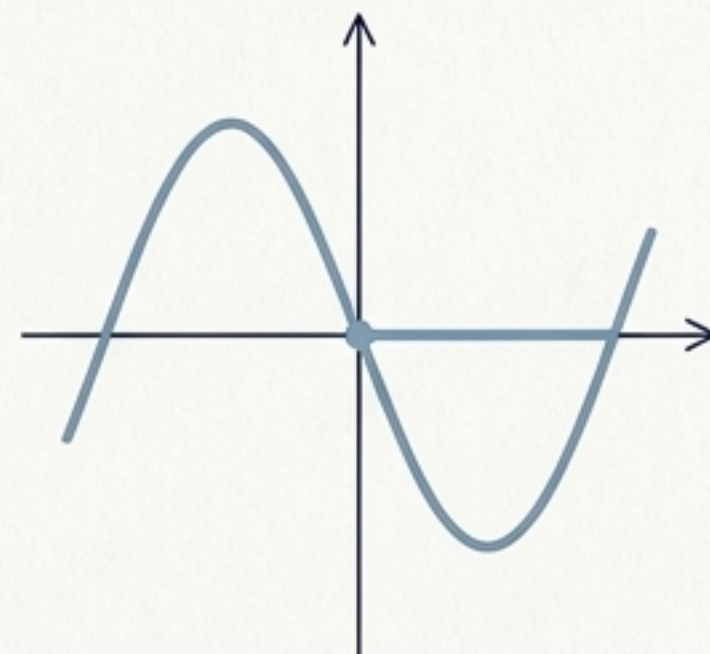
The Algebraic World



Algebraic Rank (r)

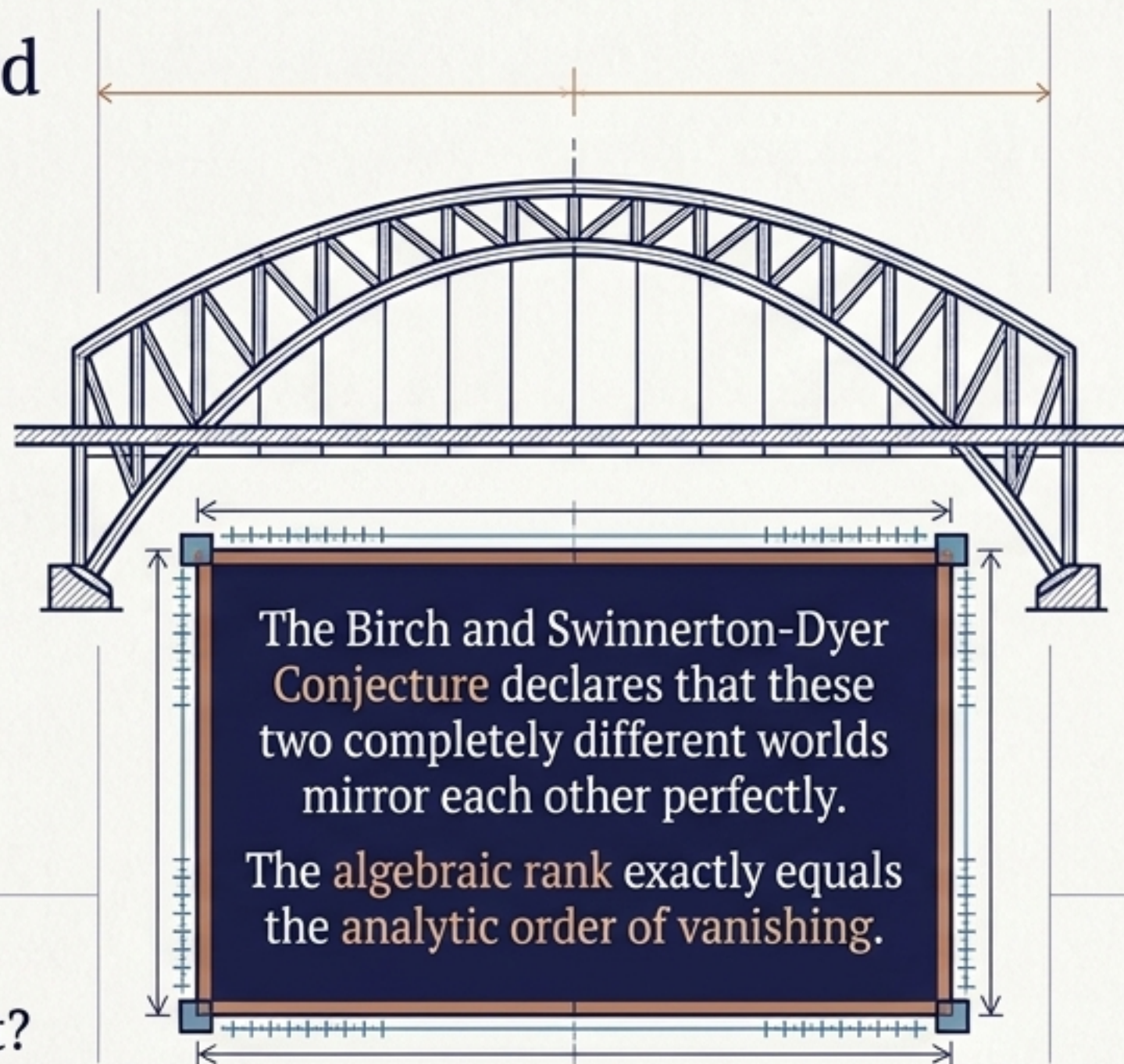
How many **independent** points of **infinite order** exist?

The Analytic World



Order of Vanishing
at $s = 1$

How **flat** does the L-function
get at its **exact center**?



Anatomy of the Exact Formula

$L^{(r)}(E,1)/r! :$

The first non-zero Taylor coefficient of the L-function at $s=1$.

Omega :

Real periods (the actual geometric area of the curve's real components).

$$\frac{L^{(r)}(E,1)}{r!} = \frac{\Omega \cdot \text{Reg} \cdot \#\text{Sha} \cdot \text{Product}(c_p)}{(\#\text{torsion})^2}$$

Reg :

Regulator (A matrix measuring the physical heights and complexity of the rational points).

#torsion :

The finite loop of points (from the Rank 0 concept).

Product(c_p) :

Tamagawa Numbers (Counts of bad reduction points at specific prime intersections).

#Sha :

The Tate-Shafarevich Group. The deepest mystery. A group measuring invisible solutions that fail to bridge local prime math and global rational math.

The Numerical Evidence: Curve 37a1

Analytic Output

$$L'(E, 1) = 0.30599977$$

Algebraic Inputs

$$\text{Area} \times \text{Regulator} = 5.98691729$$

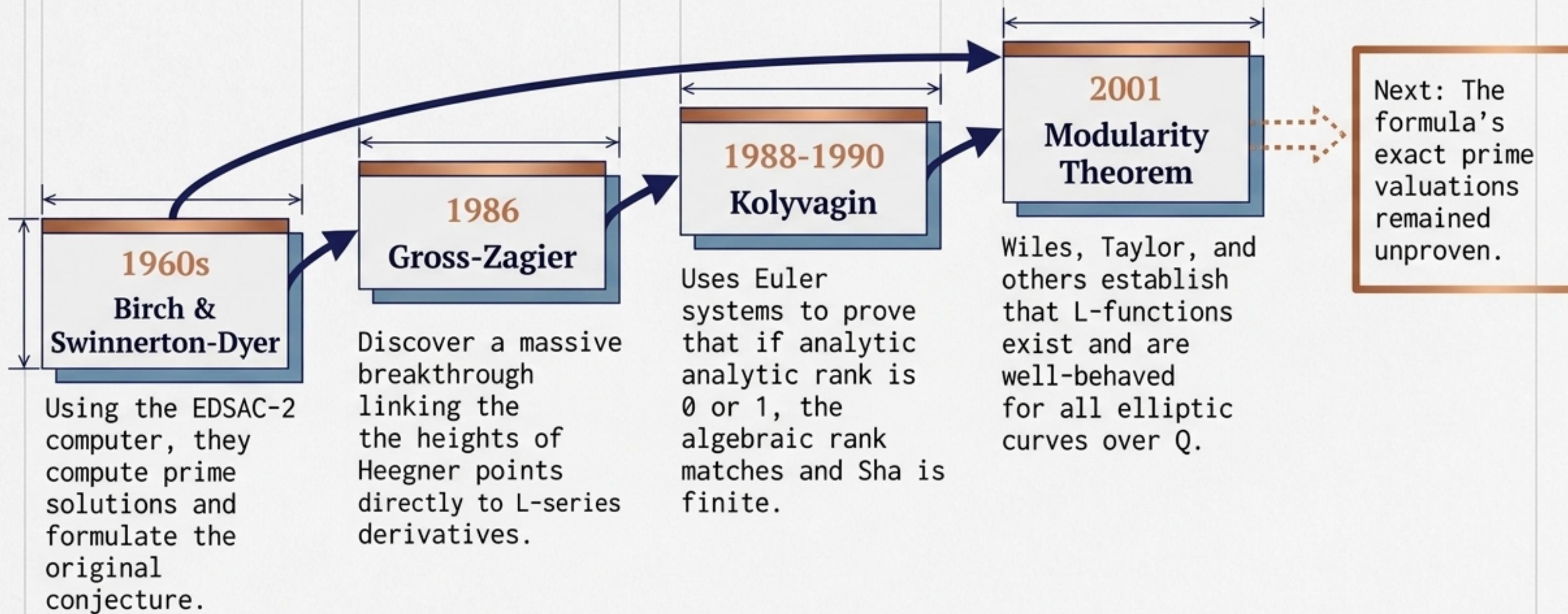
$$\text{Torsion factor} = 0.05111141$$

$$\text{Sha} = 1 \text{ (Trivial)}$$

$$5.98691729 \times 0.05111141 \times 1 = 0.30599977$$

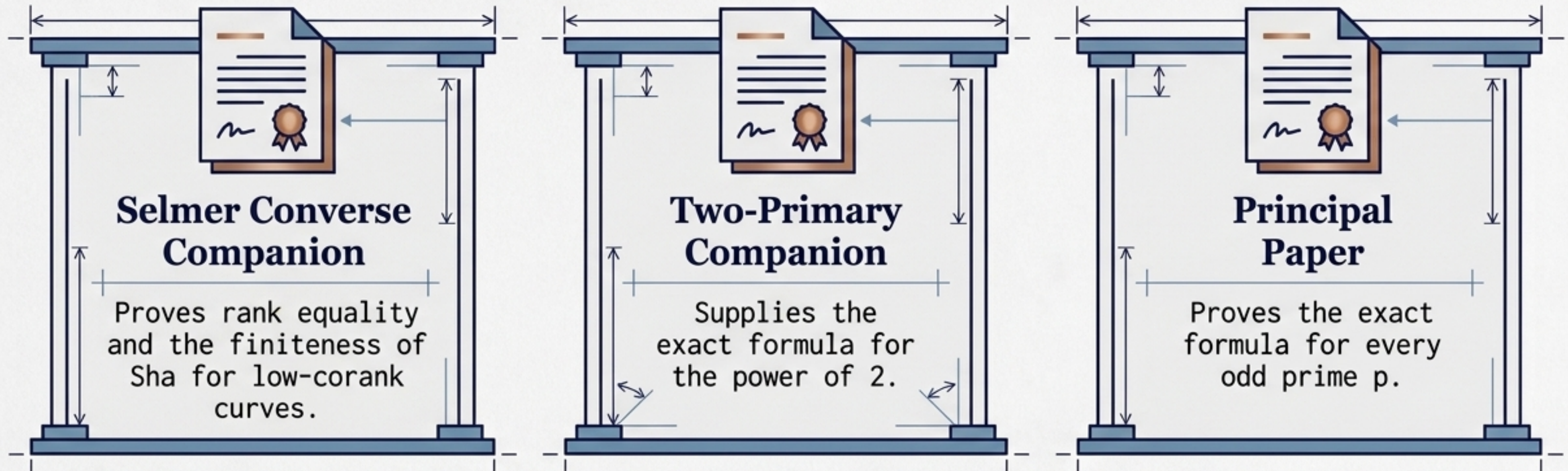
The exact formula works to the final decimal. But proving this holds true for EVERY elliptic curve has been one of the Millennium Prize Problems for over 60 years.

A Century-Long Bridge: The Road to 2026



The October 2026 Breakthrough

On October 3, 2026, an OpenAI internal model produced a catalog of preprints proving the exact BSD formula for a vast class of curves.



The Scope: The proof covers every elliptic curve over $\mathbb{Q}$ whose q -Selmer corank is 0 or 1 for some prime q (which perfectly aligns with analytic rank 0 or 1).

The Proof Mechanism: The Discrepancy Metric

The Concept

To prove the formula is exact, the AI established a discrepancy metric:

$$X_p(E) = v_p(Q_E) - v_p(\#Sha).$$

If BSD is true, this metric must be exactly zero.

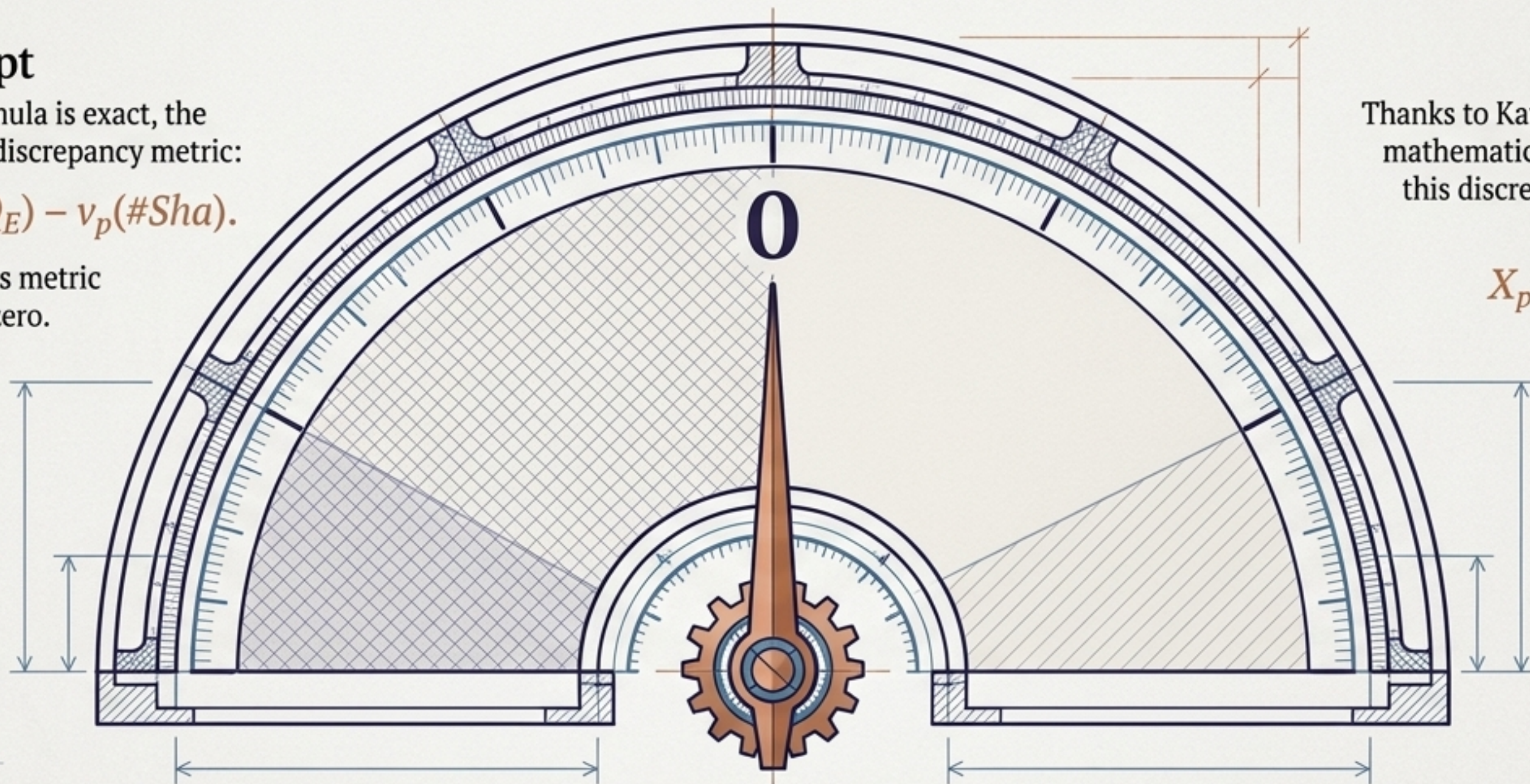
The Floor

Thanks to Kato's zeta elements, mathematicians already knew this discrepancy could never fall below zero:

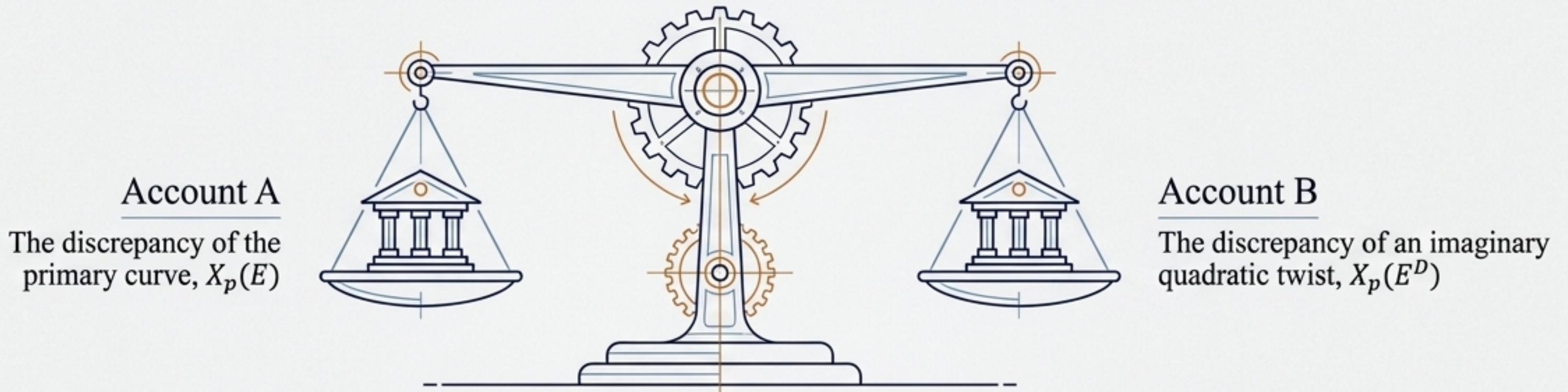
$$X_p(E) \geq 0.$$

The Challenge

The discrepancy is positive or zero.
How do you force it to be exactly zero?



The Two Bank Accounts Metaphor



Step 1: The Identity.

A Heegner point identity and a unit comparison of power series lock these two curves together, proving their sum is exactly zero: $X_p(E) + X_p(E^D) = 0$.

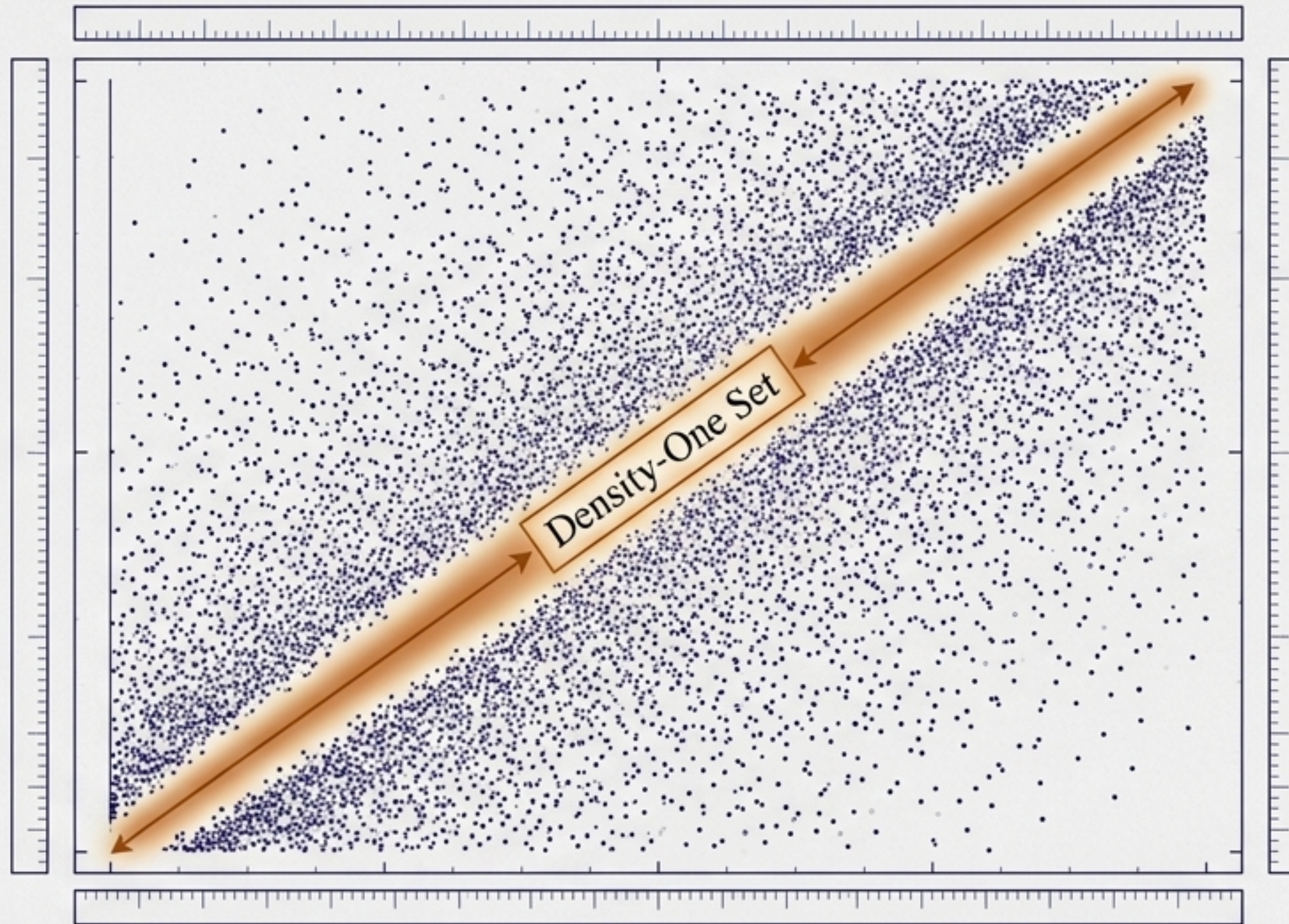
Step 2: The Logic.

Like two bank accounts that CANNOT be overdrawn (thanks to the ≥ 0 floor), if their total combined balance is exactly zero, both accounts must hold exactly nothing.
The discrepancy is zero. The formula is exact.

Expanding the Scope: Family 006

The Application

The proof does not just work for isolated examples. Using a framework called Family 006, the exact formula is shown to hold for a density-one set of quadratic twists for every curve.



What This Means

For this vast infinity of curves, the analytic rank equals the Mordell–Weil rank, and the mysterious Sha group is definitively proven to be finite. The Rosetta stone translates perfectly.

Synthesis: The Edge of Knowledge

Solved



Curves with Analytic Rank 0.

Curves with Analytic Rank 1.

Elliptic curves strictly over the field of rational numbers ($\mathbb{Q}$).

The Frontier

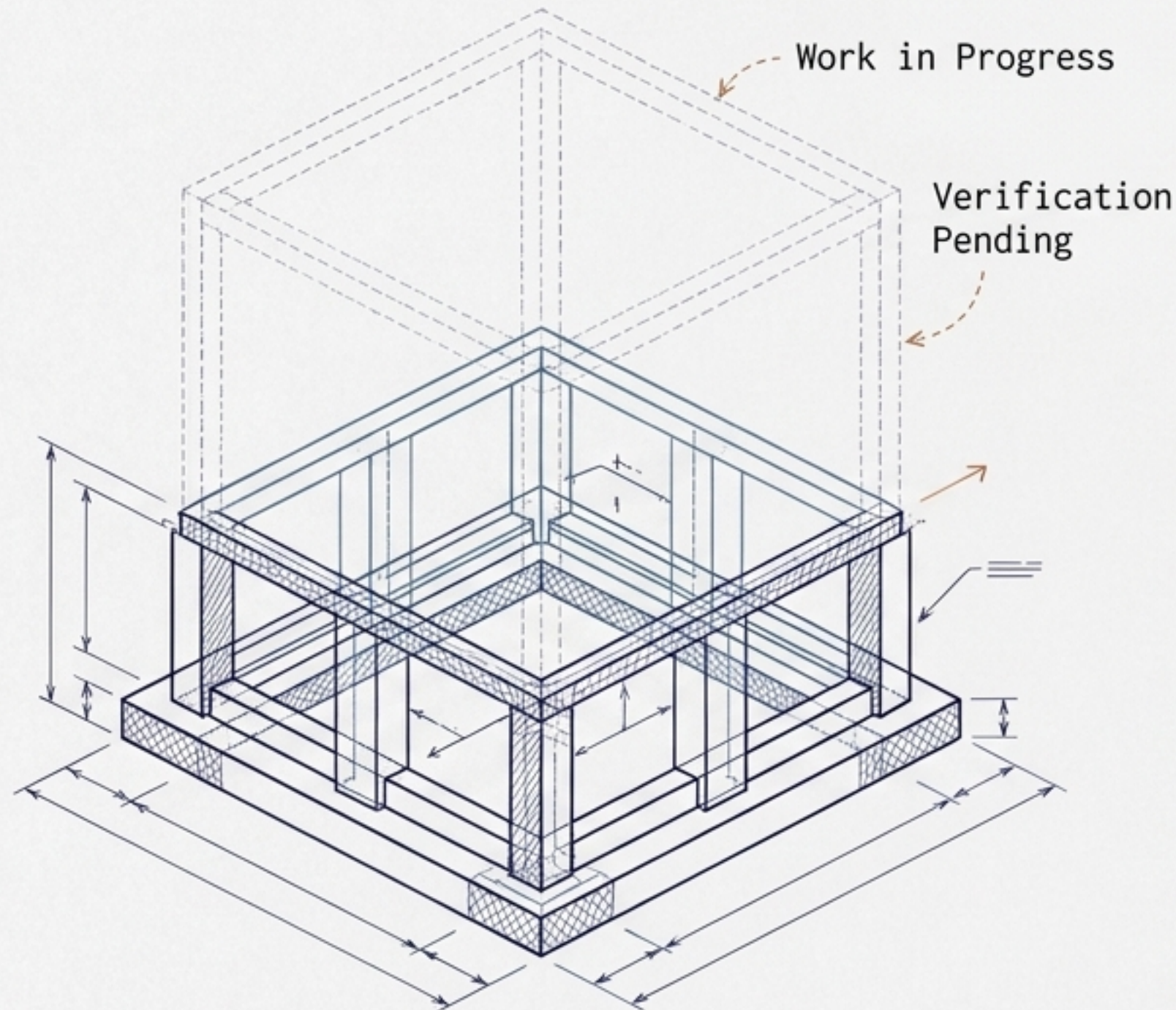


Curves with Analytic Rank 2 or higher (where Heegner points cannot directly reach).

Elliptic curves over other, more complex number fields.

Current Boundary of Mathematical Proof

The Final Caveat: Trust vs. Verification



1 The State of the Art

This breakthrough exists as an AI-produced preprint. The mathematical logic is groundbreaking and highly detailed.

2 The Missing Link

The proof currently exists without Lean formalization. Lean is the machine-readable programming language mathematicians use to computationally verify proofs line-by-line.

3 Takeaway

The logic bridges a century-old gap, but the ultimate, absolute verification is still pending. The blueprint is drawn; the foundation is awaiting its final inspection.